DOWE FLATS

UPDATED WETLAND INVENTORY REPORT



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1.0 INTRODUCTION

This report follows a request by Mr. John Lohr, Plant Manager of Southwestern Portland Cement Company, to review and update the wetland inventory for the proposed Dowe Flats mine project. The original wetland inventory was completed in October 1987 and documented in a report titled "Wetland Assessment: Dowe Flats and Foothills Reservoir Properties". A number of additions to the inventory were initiated and completed during the period of 1990 through 1993 to reflect modifications to the mine area boundaries and changes in the wetland boundary configurations. This report intends to update and document the status of the wetland inventory of the entire project area as of January 1994.

2.0 SITE DESCRIPTION

The subject property is located approximately 2.5 miles east of the Town of Lyons north of Colorado State Highway 66. Limits of the wetland study area correspond to the proposed mine permit area boundary plus the linkage of the haul road to the existing Cement Plant site south of Highway 66 (Figure 1). The study area encompass approximately three square miles.

The majority of the site is currently being used for cattle grazing and other agricultural purposes including wheat, corn and alfalfa production. Several small intermittent drainages occur throughout the area and generally drain from north to south. The St. Vrain Supply Canal, a large water delivery conduit, traverses around the project area along the western and northern boundaries. Four large irrigation canals cross the area from west to east near the southern project boundary. St. Vrain Creek crosses through the southernmost extent of the project just north of the Cement Plant. Rabbit Mountain, a 1,119 acre Boulder County open space parcel, abuts the northeastern corner of the project area.

3.0 STUDY PURPOSE

The purpose of conducting and updating the wetland inventory was to identify the current location and extent of existing wetland resources. Since the 1977 enactment of the federal Clean Water Act, Section 404 permit program, the protection of wetlands and waters has become an issue of major national concern. The Section 404 permitting process regulates the discharge of dredged or fill material into all waters of the United States, including streams, lakes and wetlands. The program is jointly administered and enforced by the U.S. Army Corps of Engineers (Corps) and the U.S. Environmental Protection Agency (EPA). The placement of fill in aquatic ecosystems constitutes a direct environmental impact and must be permitted in advance by the Corps.

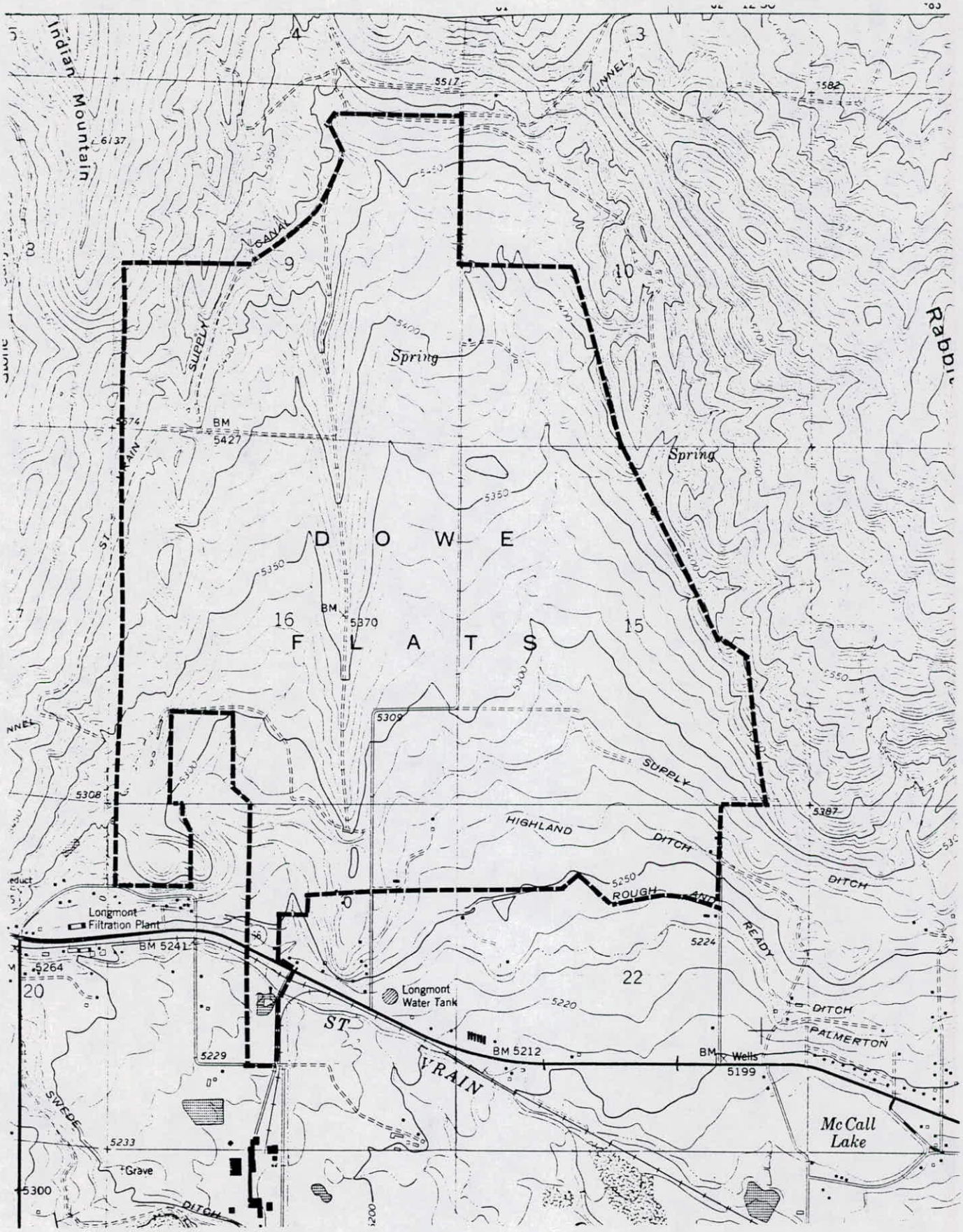


Figure 1: Boundaries of study area (illustrated in dashed line).

Knowledge of the locations of existing wetlands and their regulatory status will: 1) alert mine engineers and planners to specific areas where mine development should be avoided and/or minimized, 2) provide information that would be required for the submittal of a Section 404 permit application if wetlands are to be modified by mine development, and 3) supplement other environmental data collected for the site.

4.0 METHODS

4.1 In-House Data Gathering

Topographic mapping was produced at a scale of 1"=200' for the entire project area and served as the base upon which field identified wetlands were recorded. An aerial overflight was performed on October 23, 1993, yielding a color aerial photograph enlarged to a scale of 1"=500'. This aerial was used in the office to verify and confirm the field collected data.

4.2 Field Identification

The wetland definition that guided this study is published in the Federal Register as part of the Clean Water Act of 1977 (33 CFR Section 323.2), defining wetlands as:

> "..those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and under normal conditions do support, a prevalence of vegetation typically adapted for life in saturated soil conditions. Wetlands generally include swamps, marshes, bogs, and similar areas."

Methods described in the Corps of Engineers Wetland Delineation Manual (1987) were used to delineate wetland boundaries. This method requires the presence of positive wetland characteristics from each of the three parameters of vegetation, soils and hydrology.

Hydrophytic vegetation, hydric soils, and wetland hydrologic data collected on-site was used to differentiate between wetland and non-wetland areas. For the purpose of this study, vegetation was identified according to Weber (1990) and soil color data was obtained from Munsell (1990). Wetland community descriptions are found in Cowardin et al. (1979).

Areas determined to be wetlands were mapped in the field using prominent landmarks and a measuring wheel, and later confirmed from the October 1993 color aerial photograph.

4.3 Map Production

Field delineated wetlands were illustrated and recorded on the 1"=200' scale topographic maps (Appendix I - 2 sheets). Wetland surface areas were planimetered directly from the maps to determine their size.

4.4 Corps of Engineer Field Verification

Mr. Terry McKee, Environmental Specialist with the U.S. Army Corps of Engineers, visited the site on November 10, 1993, to review the updated wetland boundaries. All of his field observations and revisions have been incorporated into the set of final maps contained in Appendix I. A set of the final maps have been forwarded to the Corps' office with a request for a final letter of concurrence. Upon issuance of the Corps' letter, the inventory will be valid for a period of three years.

5.0 RESULTS

5.1 Wetland Acreage

A total of 70.18 acres of wetlands were inventoried within the limits of the project area (Table 1). A total of 34.06 acres occur in the east valley (east of Limestone Ridge), 22.25 in the west valley (west of Limestone Ridge), 10.05 occur along and between the four irrigation ditches in the southern portion of the project, and 3.82 occur south of Colorado State Highway 66 and north of the Cement Plant.

5.2 Wetland Description

5.2.1 East Valley

The majority of the wetlands found within the east valley are contained within two drainages which flow intermittently from north to south. Wetlands associated with the easternmost drainage originate downslope of the St.Vrain Supply Canal and cross under County Road 47. Another branch of the drainage continues to the north into the Rabbit Mountain open space area. A number of small impoundments occur along the length of the drainage, presumably to encourage water storage for agricultural use.

The second drainage flows west of and parallel to the first. Wetlands associated with the drainage also originate downslope of the St. Vrain Supply Canal. Contribution

Table 1: Summary of Wetland Acreages by Location

LOCATION/NUMBER	SIZE (In acres)
West Valley	**West Valley**
No. 5	0.22
No. 6	2.09
No. 7	0.33
No. 8	3.69
No. 14	0.48
No. 15	0.38
No. 16	0.15
No. 17	6.63
No. 18	3.66
No. 20	1.20
No. 21	0.31
No. 22	1.20
No. 23	1.13
No. 24	0.15
No. 32	0.40
No. 33	0.23
Subtotal:	**22.25**
East Valley	**East Valley**
No. 1	3.83
No. 2	0.22
No. 3	15.63
No. 4	0.39
No. 9	5.99
No. 10	1.99
No. 11	0.63
No. 12	0.63
No. 13	4.69
No. 19	0.06
Subtotal:	**34.06**
Irrigation Ditches	**Irrigation Ditches**
No. 25	0.53
No. 26	2.88
No. 27	0.04
No. 28	0.04
No. 29	2.00
No. 30	1.34
No. 31	3.22
Subtotal:	**10.05**

(continued) Table 1: Summary of Wetland Acreages by Location

S. of Hwy. 66	S. of Hwy. 66
No. 34	0.29
No. 35	0.56
No. 36	0.19
No. 37	0.13
No. 38	0.09
No. 39	2.56
Subtotal:	**3.82**

Grand Total:	**70.18**

of wetland hydrologic support from the canal has not been determined, although much of the wetland area is not visible or is reduced in size based on 1949 black and white photography taken prior to canal construction.

Both drainages have been historically modified for irrigation conveyance and are currently maintained for that purpose. Lateral ditches connect the drainages and divert surface water into adjacent fields.

Cattail (*Typha spp.*) vegetation occurs in the impounded areas along with bulrush (*Schoenoplectus lacustris*) and other obligate wetland species. Wetland forbs and grasses (*Schoenoplectus pungens, Carex spp., Juncus spp., Phalaroides arundinaceae*) dominate the remainder of the drainageway. An occasional peach-leaved willow (*Salix amygdaloides*) and scattered sandbar willow (*Salix exigua*) occur throughout, but are more abundant in the upper portions of the drainageways.

5.2.2 West Valley

Several minor drainageways and swales are located in the west valley. These wetlands either contain an intermittent surface drainage or are supported by subsurface groundwater. Each arises in close proximity to the St. Vrain Supply Canal, suggesting some level of hydrologic support from the canal. Historic existence of these wetlands prior to canal construction is unclear, although many of the wet areas are not visible or are reduced in size based on the 1949 black and white photography. Wetland boundaries in the west valley may be considered dynamic in nature due to the apparent hydrologic influence of the St. Vrain Supply Canal.

Evidence of irrigation modification in the form of small impoundments and lateral ditches is apparent in many of the drainages and swales.

The vegetative composition of the wetlands in the west valley is similar to that described for the east valley.

5.2.3 Irrigation Ditches

Seven isolated wetlands areas are found amongst the four irrigation ditches which flow from west to east across the southern portion of the project. All are grazed quite heavily and exhibit sedge (*Carex spp.*) and three-square (*Schoenoplectus pungens*) as wetland vegetative indicators. The riparian vegetation of cottonwoods (*Populus spp.*) and willows (*Salix spp.*) found immediately adjacent and within the ditch banks does not qualify as jurisdictional wetland.

5.2.4 Wetlands South of Highway 66

Immediately south of the highway in the borrow ditch and in the borrow ditch of the railroad, cattail wetlands are found in long, narrow strips. No wetlands were found in the riparian gallery forest of the St. Vrain River, except within the active channel area where gravel bars have temporarily revegetated with sandbar willow. A few remnant floodplain wetlands dominated by the wetland forb and grass species identified earlier are found immediately north of the plant in an old oxbow feature, and just north of the gravel pit ponds.

5.3 Wetland Classification

Wetlands can be classified for purposes of nationwide inventory, evaluation and management using the U.S. Fish and Wildlife Service (USFWS) classification system (Cowardin et al., 1979). The structure of classification is hierarchical, progressing from systems and subsystems at the general level, through more specific levels of classes, subclasses, and dominance types.

Wetlands within the survey area can be classified within the palustrine wetland system. The palustrine system describes wetlands which are dominated by trees, shrubs, emergent plants, and emergent mosses or lichens. These wetlands can further be categorized within the emergent wetland class. Emergent wetlands include areas dominated by grasses and forbs.

Classification of the project area wetlands can be outlined as follows:

SYSTEM:	Palustrine
SUBSYSTEM:	None
CLASS:	emergent wetland
SUBCLASS:	persistent
DOMINANCE TYPES:	*Carex spp., Juncus spp., Typha spp., Schoenoplectus pungens, Phalaroides arundinaceae*
WATER REGIME:	seasonally saturated
WATER CHEMISTRY:	fresh
SOIL:	mineral

6.0 DISCUSSION

6.1 Federal Regulations

The Corps will assume regulatory jurisdiction over all areas preliminarily delineated by AWC as wetland. In addition, the St. Vrain River will regulated as a waters of the U.S. The Corps will not extend jurisdiction to the irrigation ditches or the St. Vrain Supply Canal.

A Section 404 permit will be required to place fill in regulated areas.

6.2 Permitting

Under Section 404(b), Congress empowered the Secretary of the U.S. Army, acting through the Corps of Engineers, to issue permits based on the application of guidelines developed by the EPA administrator in conjunction with the Corps. The EPA, under Section 404(c) is vested with veto power for discharges having "an unacceptably adverse effect on municipal water supplies, shellfish beds and fishing areas...wildlife or recreation areas." In addition, the EPA has enforcement authority under Section 309 for discharges without a permit, and often comments on proposed Corps permits, especially the interpretation and application of the Section 404(b)(1) guidelines.

When a development activity involves the discharge of dredged or fill material covered under Section 404, the Corps is required to ensure compliance with the EPA Section 404(b)(1) guidelines. The guidelines are based on the assumption that wetland sites should not be developed if alternative sites are available. This assumption applies even more directly if proposed development activities are not water dependent (i.e., do not require placement in an aquatic site in order to fulfill the basic project purpose).

6.2.1 Practicable Alternatives

The "practicable alternatives" section of the 404(b)(1) guidelines assumes that, if a project is non-water dependent, alternative sites that do not involve disturbance of wetland or other aquatic features, are available unless clearly demonstrated otherwise (40 CFR Section 230.10 (a)(3)). To satisfy the "practicable alternatives" test, the applicant must first prove that alternative sites would not be feasible or achieve their projects' basic purposes. The guidelines state that "what is practicable depends on cost, technical and logistic factors" and that an applicant must "consider those alternatives which are reasonable in terms of the overall scope/cost of the proposed project." However, even though an alternative site does not meet all of the desired specifications, does not maximize profit, or is somehow undesirable to the applicant, it may still be deemed feasible or practicable by the EPA.

6.2.2 Avoidance

Once it has been determined that a project is or is not water-dependent, and that no other practicable alternatives exist, permit approval depends on a determination of whether adverse impacts have been avoided to the extent possible. The process of mitigation of environmental impacts must first address the possibility of total avoidance of impacts. If impacts are deemed by the Corps and the EPA to be unavoidable, the issue of minimization of impacts will be considered and the process of development of mitigation alternatives can begin.

6.2.3 Minimization

Once it has been determined that wetlands have been avoided to the maximum extent possible, the Corps will require that wetland impacts be minimized. This means that roads must be situated in the least damaging alignments, mine excavation and stockpile sites must be placed outside of wetlands, and that other construction activities be located in the least damaging locations.

6.2.4 Mitigation

Compensatory mitigation of unavoidable wetland impacts is required to insure no net loss of wetland resources by project implementation. Mitigation can include wetland creation, restoration, or enhancement. Mitigation to impact ratios of 1.5:1 to 2:1 are common. When reviewing mitigation proposals, the Corps first requires consideration of all on-site mitigation possibilities. If it is deemed that on-site options are not available or feasible, off-site compensation will be considered.

6.2.5 Nationwide Permits

The nationwide permitting program contains thirty-seven permits which pre-authorize routine activities occurring in wetlands or waters of the U.S. Minor road crossings of drainages are authorized by nationwide permit #14. Conditions of this permit include: 1) that the crossing be bridged, culverted, or otherwise designed to avoid restriction of expected high flows, 2) that the associated fill does not extend into wetlands beyond 200 feet combined on either side of the annual high water elevation, and 3) that no more than 1/3 of a surface acre of wetland or water is disturbed by the construction action. Mitigation is suggested for wetland impacts. As with all Section 404 permits, avoidance and minimization must be considered.

A nationwide #14 permit authorization has been issued for the haul road crossing of the St. Vrain River (permit authorization No. 199380521 dated 9 August 1993).

Nationwide permit authorizations are valid for up to five years or until they are modified or revoked by Congress (reviews occur every five years).

6.2.6 Individual Permits

If the project cannot be developed using nationwide permits, then project review will occur through an individual permit application. All individual permit applications are advertised for public comment and are reviewed by federal, state and local agencies. The Corps generally requires that both direct (i.e., wetland filling/excavation) and indirect (i.e., draining, etc.) impacts be disclosed for review. Compensatory mitigation is required to replace all wetlands which are filled.

Individual permits are valid for three to five years.

6.3 Recommendations

Design a mine and reclamation plan which avoids and minimizes wetland impacts to the maximum extent possible. Locate excavation pits, temporary stockpiles and haul roads outside of wetlands wherever practicable. Design reclamation features around and to compliment existing wetlands. Incorporate created wetland habitats into the reclamation plan as possible to increase site diversity and increase wetland resources within the region.

During construction, retain a wetland delineation expert to field stake the limits of wetlands within each discrete area of mining activity. These staked areas should then be reinforced by some mechanism like temporary snowfence to indicate their protection during construction. Phase the staking process concurrent with mining activity.

The location and extent of wetlands may change over the life of the mine project given the dynamic nature of the hydrology with particular reference to the area west of County Road 47. The Corps is willing to field review and reassess any wetland areas that may change substantially during the life of the project.

7.0 LITERATURE CITED

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